

JC 2 H2 Mathematics Preliminary Examination 2009
Paper 2 -- Marking Scheme

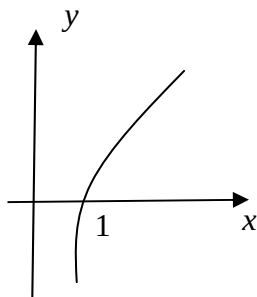
Section A: Pure Mathematics [40 marks]

1 (i) $\frac{4-r}{r(r-1)(r-2)} = \frac{A}{r} + \frac{B}{r-1} + \frac{C}{r-2}$ By Cover Up rule or any other methods, $A = 2$, $B = -3$ and $C = 1$ <i>[Subtract 1 mark for every mistake]</i> (ii) $\sum_{r=3}^n \frac{4-r}{r(r-1)(r-2)} = \sum_{r=3}^n \left(\frac{2}{r} - \frac{3}{r-1} + \frac{1}{r-2} \right)$ $= \left(\frac{2}{3} - \frac{3}{3-1} + \frac{1}{3-2} \right)$ $+ \left(\frac{2}{4} - \frac{3}{4-1} + \frac{1}{4-2} \right)$ $+ \left(\frac{2}{5} - \frac{3}{5-1} + \frac{1}{5-2} \right)$ $+ \left(\frac{2}{6} - \frac{3}{6-1} + \frac{1}{6-2} \right)$ $\vdots$ $+ \left(\frac{2}{n-2} - \frac{3}{n-3} + \frac{1}{n-4} \right)$ $+ \left(\frac{2}{n-1} - \frac{3}{n-2} + \frac{1}{n-3} \right)$ $+ \left(\frac{2}{n} - \frac{3}{n-1} + \frac{1}{n-2} \right)$ $= -\frac{3}{2} + 1 + \frac{1}{2} + \frac{2}{n-1} - \frac{3}{n-1} + \frac{2}{n}$ $= \frac{2}{n} - \frac{1}{n-1}$ $= \frac{n-2}{n(n-1)}$ (iii) $\sum_{r=1}^n \frac{2-r}{r(r+1)(r+2)} \xrightarrow{k=r+2} \sum_{k=3}^{n+2} \frac{2-(k-2)}{(k-2)(k-2+1)(k-2+2)}$ $= \sum_{k=3}^{n+2} \frac{4-k}{(k-2)(k-1)(k)}$ $= \sum_{r=3}^{n+2} \frac{4-r}{(r-2)(r-1)(r)}$	[B2] [M1] [M1] [M1] [A1] [M1]
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	$= \frac{(n+2)-2}{(n+2)((n+2)-1)}$ $= \frac{n}{(n+2)(n+1)} \quad [\text{Accept answer in partial fractions}]$	[M1] [A1]
2	(i) By G.C Intersection point (1.05395, -0.947453), (4.3919, 0.47976) Area = $\int_{1.05395}^{4.3919} \ln(x) - e^{x-4} dx$ [M1 - correct limits ; M1 - correct form] Area = 1.68 (ii) $V_x = \pi \int_0^b (x^2)^2 dx$ $V_x = \pi \left[\frac{x^5}{5} \right]_0^b = \pi \frac{b^5}{5}$ $V_y = \pi(b^2)(b^2) - \pi \int_0^{b^2} y dy$ [M1 - Vol of cylinder ; M1 - Vol of revolution abt y-axis] $V_y = \pi b^4 - \pi \left[\frac{y^2}{2} \right]_0^{b^2} = \pi \frac{b^4}{2}$ $\pi \frac{b^5}{5} = \pi \frac{b^4}{2}$ $b^4 \left(\frac{b}{5} - \frac{1}{2} \right) = 0$ $b = 0$ (rejected) $b = \frac{5}{2}$ <u>Alternative Solution</u> $V_x = \pi \int_0^b (x^2)^2 dx$ $V_x = \pi \left[\frac{x^5}{5} \right]_0^b = \pi \frac{b^5}{5}$ $V_y = \pi(b^2)(b^2) - \pi \int_1^{b^2+1} y - 1 dy$ [M1 - vol of cylinder, M1 - vol of revolution abt y-axis] $V_y = \pi b^4 - \pi \left[\frac{y^2}{2} - y \right]_1^{b^2+1} = \pi b^4 - \pi \left[\frac{(b^2+1)^2}{2} - (b^2+1) - \frac{1}{2} + 1 \right]$	[M1+M1] [A1] [M1] [M1] [M1+M1] [M1] [A1] [M1] [M1] [M1+M1]

	$V_y = \pi b^4 - \pi \left[\frac{y^2}{2} - y \right]_1^{b^2+1} = \pi b^4 - \pi \frac{b^4}{2} = \pi \frac{b^4}{2}$ $b = 0$ (rejected) or $b = \frac{5}{2}$	[M1] [A1]
3 (a)	$-8\sqrt{3} + 8i = 2^4 e^{i\frac{5\pi}{6}}$ $z^8 = 2^4 e^{i\frac{5\pi}{6}} \times e^{i2k\pi} = 2^4 e^{i\pi(2k+\frac{5}{6})}$ $z = \sqrt[8]{2} e^{i\frac{\pi}{8}(2k+\frac{5}{6})}, k = 0, \pm 1, \pm 2, \pm 3, -4$ (b) M1/2, M1/2 - Correct marking of V and W (i) B1 - correct locus B ½ if the perpendicular bisector does not pass through origin (ii) B1 - correct locus B1 - showing awareness that the halfline is parallel to $-iw$ Let Q be the point of intersection. Using vector addition, $\overrightarrow{OQ} = \overrightarrow{OW} + \overrightarrow{WQ} = w + (-iw)$ It is a square.	[M½ - mod, M½ - arg] [M1] [M1 – taking 8 th root, A1] [M1,A1] [B1]

4(i)



Since any horizontal line $y = k, k \in \mathbb{R}$ will cut the graph of f at most once, hence f is one-one. Thus, f^{-1} exists.

[M1]

(ii) Let $y = x - \frac{1}{x}$

$$x = \frac{y}{2} \pm \frac{1}{2} \sqrt{y^2 + 4}$$

[M1]

[M1]

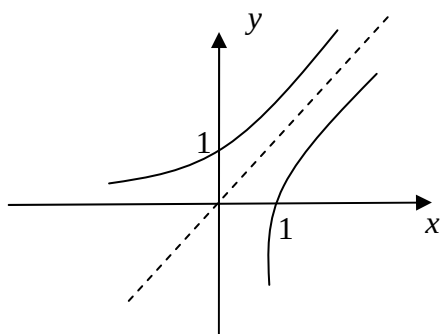
But $x > 0, \therefore x = \frac{y}{2} + \frac{1}{2} \sqrt{y^2 + 4}$

[M1]

Hence, $f^{-1} : x \rightarrow \frac{x}{2} + \frac{1}{2} \sqrt{x^2 + 4}, x \in (-\infty, \infty)$

[A½ - rule,
A½ - domain]

(iii) [B1] – correct graph of f^{-1} with $y = x$
[B1] - intercepts



(iv) $D_f : (0, \infty)$ and $R_g : [-1, 1]$.
Since $R_g \not\subseteq D_f$, therefore fg does not exist.

} [B1]

(v) Largest $D_g : (0, \pi)$

[B1]

$$fg(x) = f(\sin x)$$

$$= \sin x - \frac{1}{\sin x}$$

[M1]

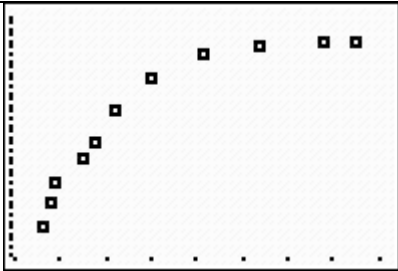
Hence, $fg : x \rightarrow \sin x - \frac{1}{\sin x}, x \in (0, \pi)$.

[A1]

Section B: Statistics [60 marks]

5(i)	- Mention of sampling frame/list. - Random starting point (i) from first 5 on the list (either explicitly or by giving an example), (ii) otherwise student must mention that once they reach the end of the list, they have to go back to the beginning of the list to pick. - Select every 5th student <i>E.g. From the school's registration list [B½], randomly select a student from the first 5 on the list as a starting point [B1]. Select every 5th student [B½] down the list from that starting point, until 300 names are obtained.</i>	[B½] [B1] [B½]
5(ii)	- A sampling frame is easily accessible. - The population can be divided into non-overlapping subgroups (e.g. boys and girls) <i>[demonstrate knowledge of subgroups being non-overlapping by explicitly mentioning or giving examples]</i>	[B1] [B1]
6	Let X be the battery life of a iPhone 3GS in hours. Unbiased est. of popn mean, $\bar{x} = \frac{\sum (x - 120)}{n} + 120 = \frac{-29}{90} + 120 = 119.6777778$ Unbiased est. of popn var, $\hat{\sigma}^2 = \frac{1}{89} \left[419 - \frac{(-29)^2}{90} \right] = 4.602871411$ $\bar{X} \sim N\left(120, \frac{4.602871411}{90}\right)$ $H_0 : \mu = 120$ $H_1 : \mu < 120$ Test statistic, $z = \frac{119.6777778 - 120}{\sqrt{\frac{4.602871411}{90}}}$ Use GC, p-value = 0.0771035664 = 0.0771 [accept 0.0771 – 0.0785] Since p-value = 0.0771 > 0.05, we do not reject H_0. Hence, there is insufficient evidence at 5% level to indicate that the mean battery life of a iPhone is less than 120 hours. Probability of concluding the mean battery life of a iPhone is less than 120 hours when it is actually 120 hours is 0.05.	[M1] [M1] [B1] [M1] [A1] [B½] [A½] [B1]

7	(a) $\frac{1}{4} = \frac{P(B \cap A')}{1 - \frac{1}{4}} \therefore P(B \cap A') = \frac{3}{16}$ $P(B \cap A) = P(B) - P(B \cap A') = \frac{1}{3} - \frac{3}{16} = \frac{7}{48} \neq 0$ Hence A and B are not mutually exclusive. (bi) $P(\text{score is } 5) = P[(3,2), (1,1,3), (1,2,2), (1,3,1), (2,1,2), (2,2,1)]$ $= \frac{1}{36} + \frac{5}{216} = \frac{11}{216}$ (bii) $P(\text{score } 3) = P[(1,1,1)] = \frac{1}{216}$ $P(\text{score } 4) = P[(3,1), (1,1,2), (1,2,1), (2,1,1)] = \frac{9}{216}$ $P(\text{score } 4 \text{score} \leq 5) = P\left(\frac{\text{score } 4 \cap \text{score} \leq 5}{\text{score} \leq 5}\right)$ $= P\left(\frac{\text{score } 4}{\text{score} \leq 5}\right)$ $= \frac{9}{1+9+11} = \frac{3}{7}$	[M1] – 1st stmt [M1] – “≠ 0” [A1] [M1] [A1] B1 for finding P(score 3, 4) [M1] [A1]
8	(a)(i) $10C4 = 210$ (ii) $10C4 - 3C2 = 207$ (b)(i) $2 \times 2 \times 2 \times (7-1)! \times 10 = 57600$ [M1 - considering permutation of wife and husband, M1 - (7-1)! × 10] (ii) $(10-1)! \times 10 = 3628800$	[B1] [M1, A1] [M1, M1, A1] [M1, A1]

9	(i) Let r.v. X_r be the number of respondents who favour greater consumer protection out of r respondents. $X_{30} \sim B(30, 0.95)$ $P(X_{30} = 25) = 0.0124$ (ii) $X_n \sim B(n, 0.95)$ $P(X_n \geq 40) > 0.98$ $1 - P(X_n \leq 39) > 0.98$ Using GC, least $n = 46$ (iii) Let r.v. Y be the number of respondents who do not favour greater consumer protection out of 60 respondents. $Y \sim B(60, 0.05)$ $Y \sim P_o(3)$ approximately. $P(X_{60} \leq 58) = P(Y \geq 2)$ $= 1 - P(Y \leq 1)$ $= 0.801$	[M1] - distn [A1] [M1] [M1] [A1] [M1] [B1] [M1] [A1]
10	(i)  Shape Label axes and 2 end points (6.6,119) & (13.5,141) Based on the scatter diagram, the linear model is not suitable even though $r (= 0.906)$ is quite close to 1. (ii) Choose Model (b): $y = ax^b$ Reason: The graph of $y = ax^b$ fits the scatter diagram better. : From the scatter diagram, we see that as x increases, y increases at a decreasing rate. (iii) $r = 0.932$ $y = ax^b$ $\ln y = \ln(ax^b)$ $\ln y = \ln a + b \ln x$ $\ln y = 4.1912 + 0.3056 \ln x$ $\ln a = 4.1912 \Rightarrow a = 66.1$ $b = 0.306$ <pre> LinReg y=ax+b a=.3055728916 b=4.191193497 r²=.8692531131 r=.9323374459 </pre> (iv) when $y = 110$, $x = 5.29$. Extrapolation hence not reliable.	[B1] [B1] [B1] [B1] [B1 – any 1 reason] [B1] [M1] [A1] [A1] [B1 + B1]

11	(a) Let A be the random variable that denotes the weight, in grams, of a bar of Brand A chocolate $A \sim N(180, 100)$ Let B be the random variable that denotes the weight, in grams, of a bar of Brand B chocolate $B \sim N(240, 400)$ $A_1 + A_2 + A_3 - 2B \sim N(60, 1900)$ $P(A_1 + A_2 + A_3 > 2B) = P(A_1 + A_2 + A_3 - 2B > 0)$ $P(A_1 + A_2 + A_3 - 2B > 0) = 0.916 \text{ (to 3 s.f.)}$ (b)(i) $\bar{A} \sim N\left(180, \frac{100}{60}\right)$ $P(\bar{A} > 179) = 0.781 \text{ (to 3 s.f.)}$ (b)(ii) No need to use Central Limit Theorem because the population follows a Normal Distribution. (c)(i) $0.02A \sim N((0.02)(180), (0.02)^2(100))$ $0.02A \sim N(3.6, 0.04)$ $P(0.02A > 3.80) = 0.1586552596 = 0.159 \text{ (to 3 s.f.)}$ (c)(ii) Let X be the random variable that denotes the number of bars of Brand A chocolate, in a box, that cost more than \$3.80. $X \sim B(100, 0.1586552596)$ $n = 100, n \geq 30$ $np = 15.86552596, np > 5$ $nq = 84.13447404, nq > 5$ $X \sim N(15.86552596, 13.34837682) \text{ approx}$ $P(X > 5) \approx P(X > 5.5)$ $P(X > 5.5) = 0.9977237843 = 0.998 \text{ (to 3 s.f.)}$	[M1] [M1] [A1] [M1] [A1] [B1] [M1] [A1] [M1] [M1] [M1] [M1] [A1]
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